

Federated Biobank Analytics

Agent orchestration across three data sites

NVFlare for distributed execution

Codex for analysis planning and development

Hackathon prototype

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The analysis problem

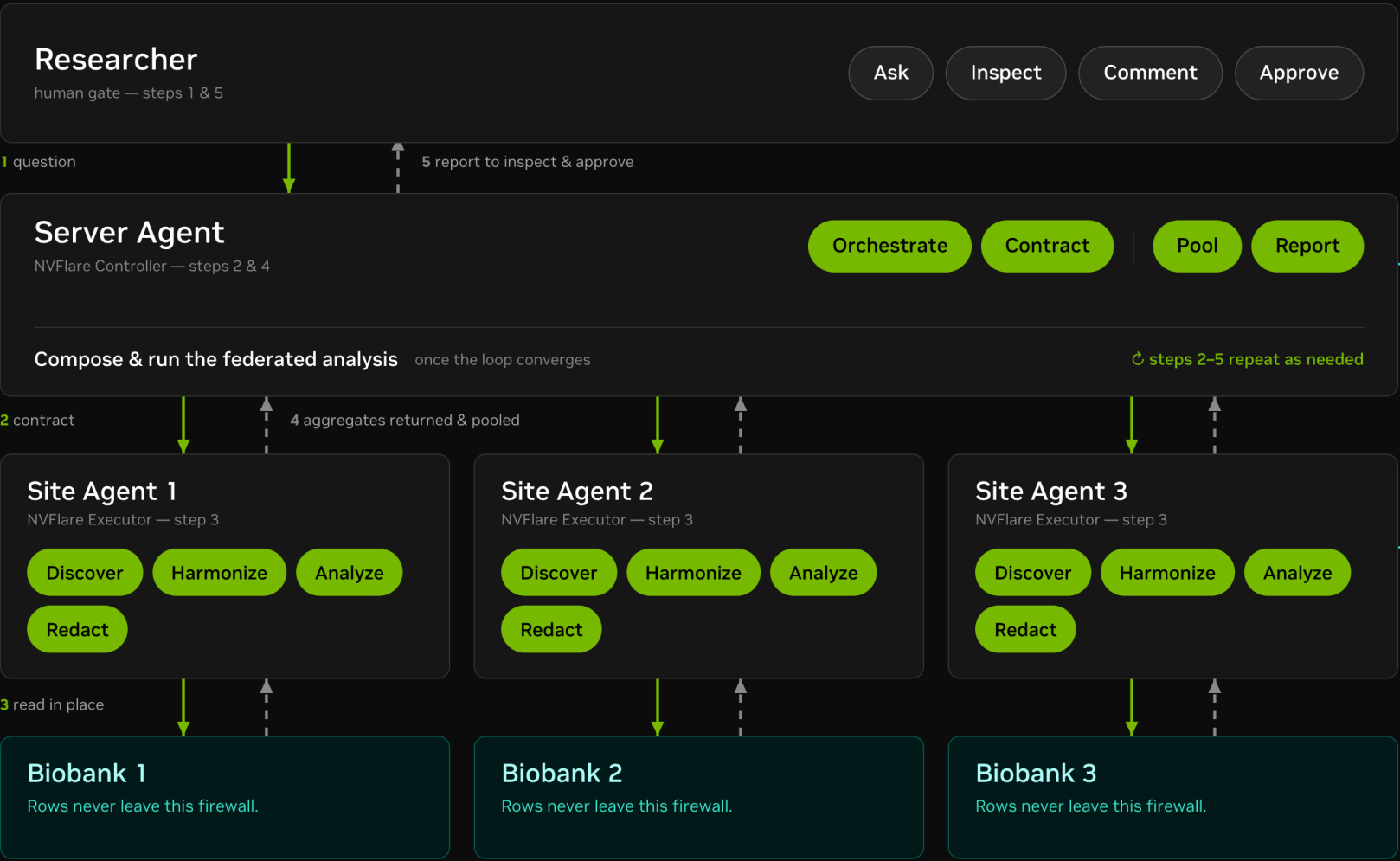
A researcher wants one answer from several biobanks, but the records cannot move to a central database.

- 01 Different schemas** Equivalent concepts may use different column names, units, or encodings.
- 02 Protected data** Patient-level rows must remain inside each institution.
- 03 Multi-site statistics** Site results require mathematically correct pooling, not simple averaging.

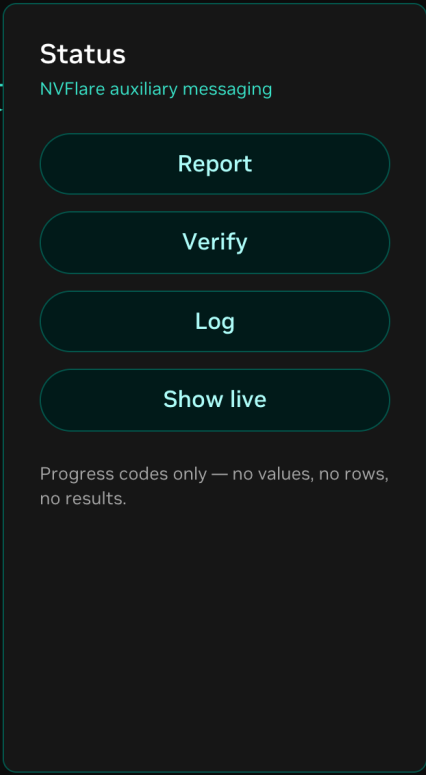
Orchestrated end to end by NVFlare

Server agent = Controller. Site agents = Executors. Data stays home. Two channels, one job: the analysis flow on the left, the UI status stream on the right.

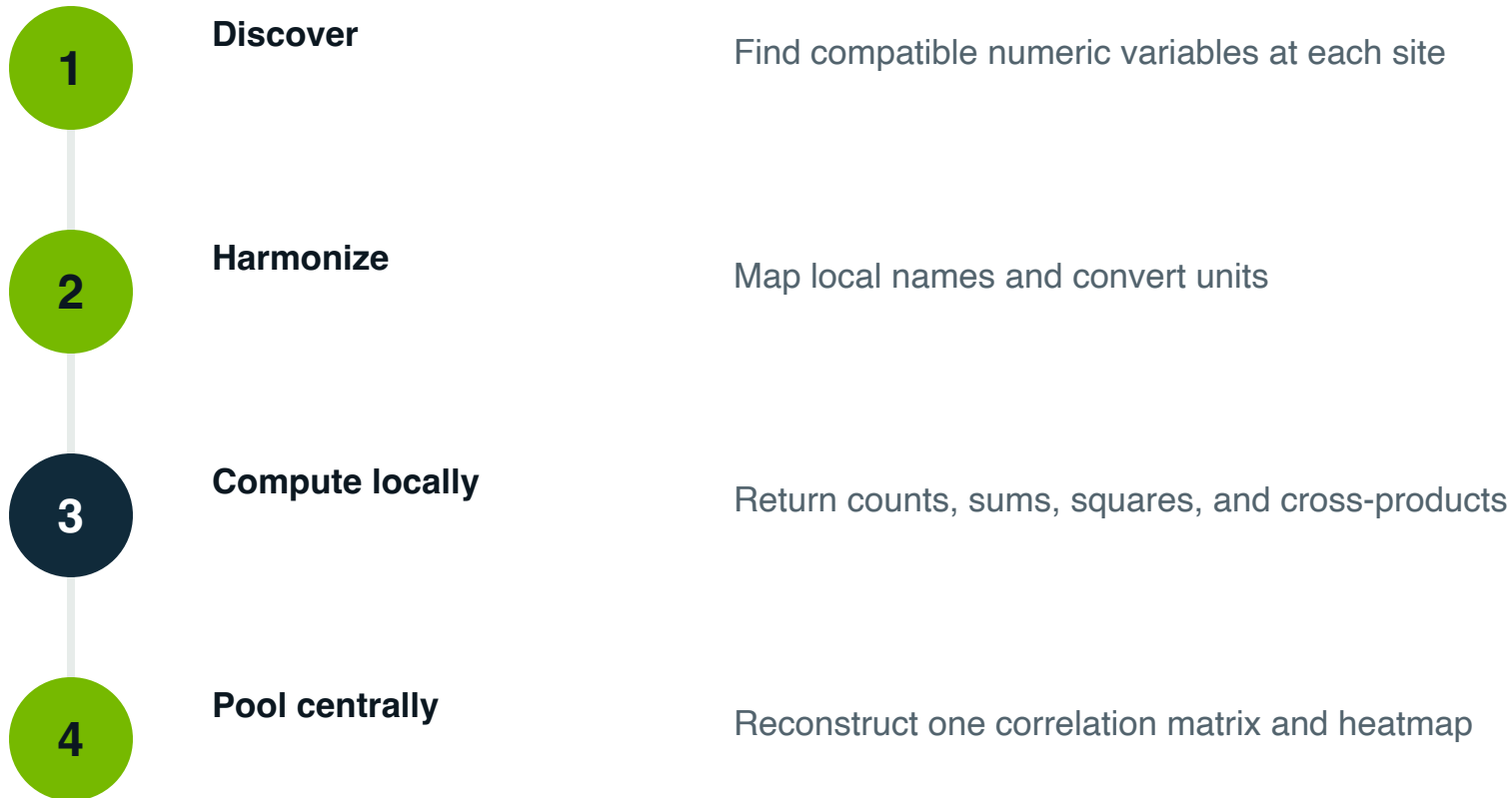
MAIN FLOW — ORCHESTRATED BY NVFLARE



UI STATUS — NVFLARE MESSAGING



Federated correlation workflow



Human and agent correlation comparison - Site A

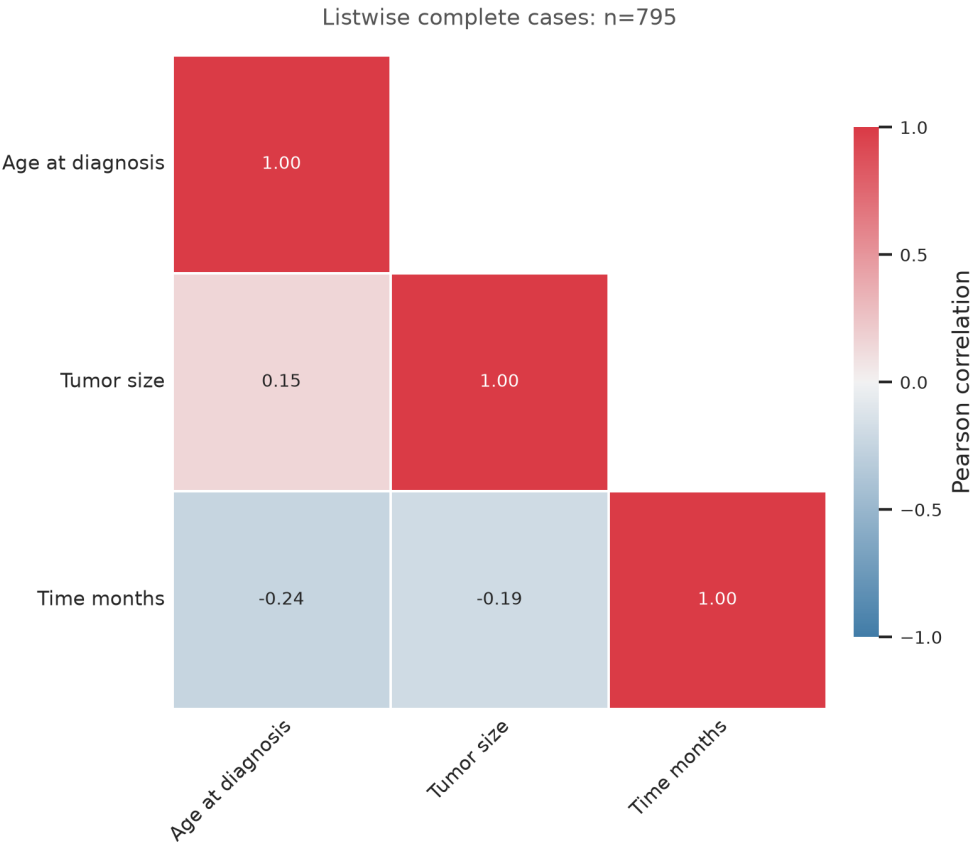
Agent analysis

Site And Federated Pearson — Site A University Hospital



Human analysis

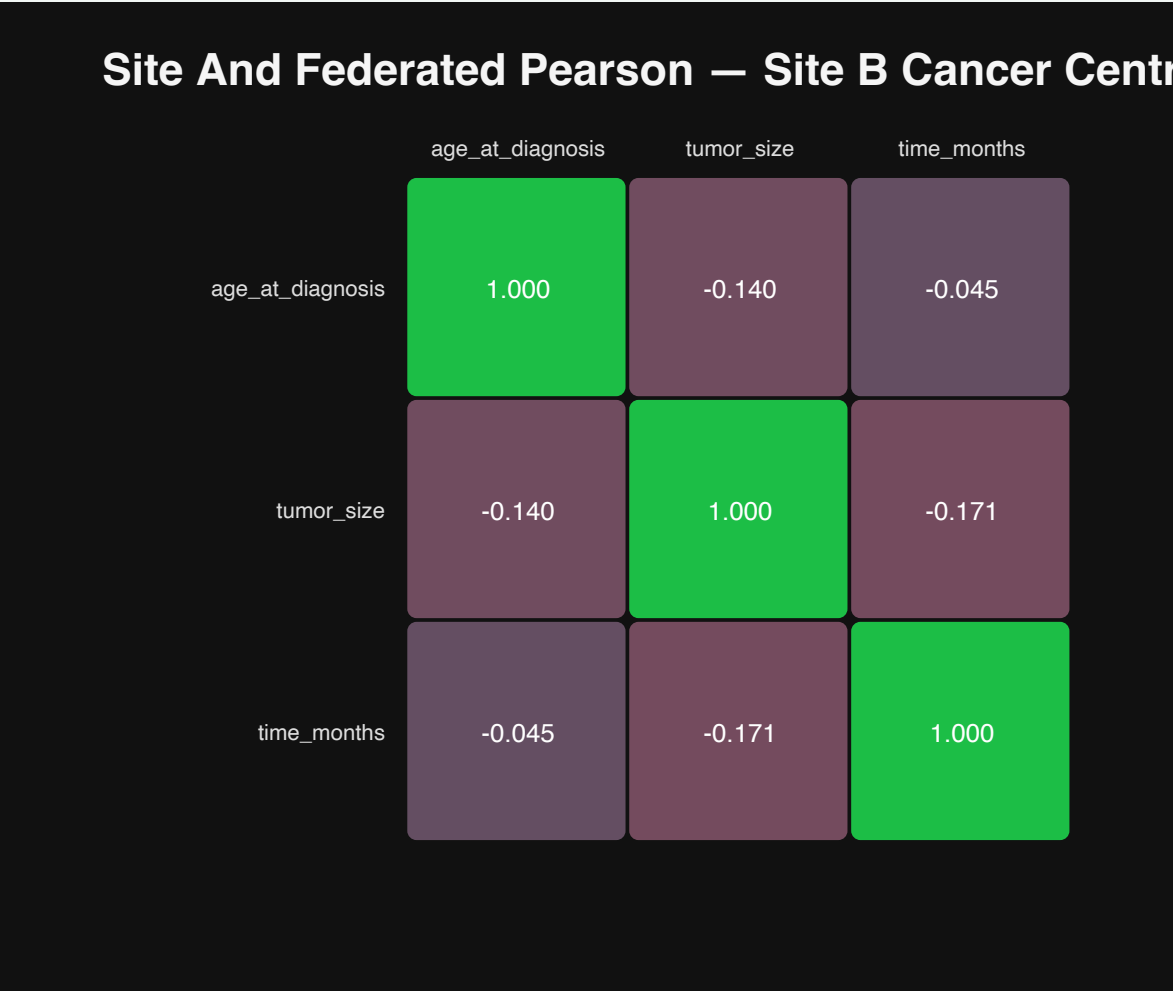
University Hospital: Local correlation heatmap



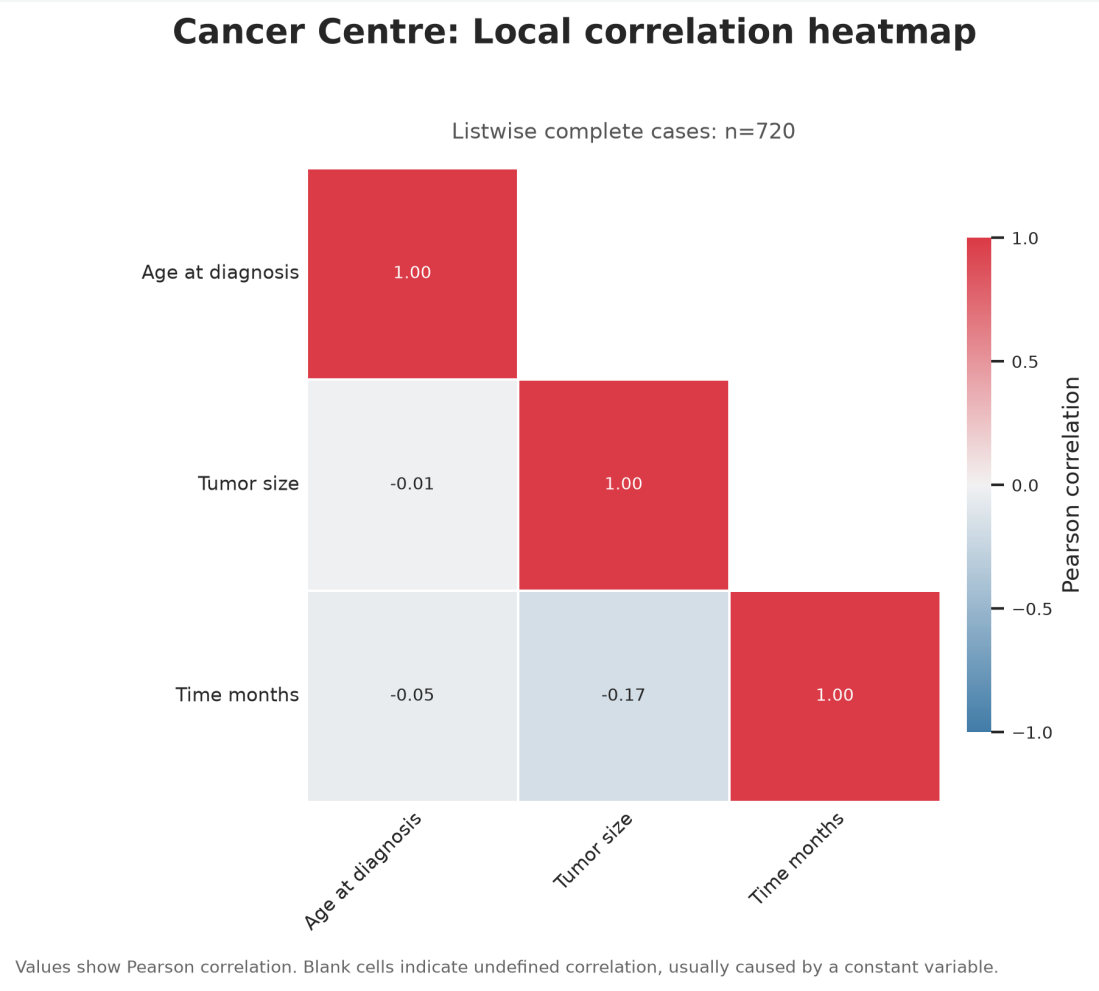
Values show Pearson correlation. Blank cells indicate undefined correlation, usually caused by a constant variable.

Human and agent correlation comparison - Site B

Agent analysis



Human analysis



Human and agent correlation comparison - Site C

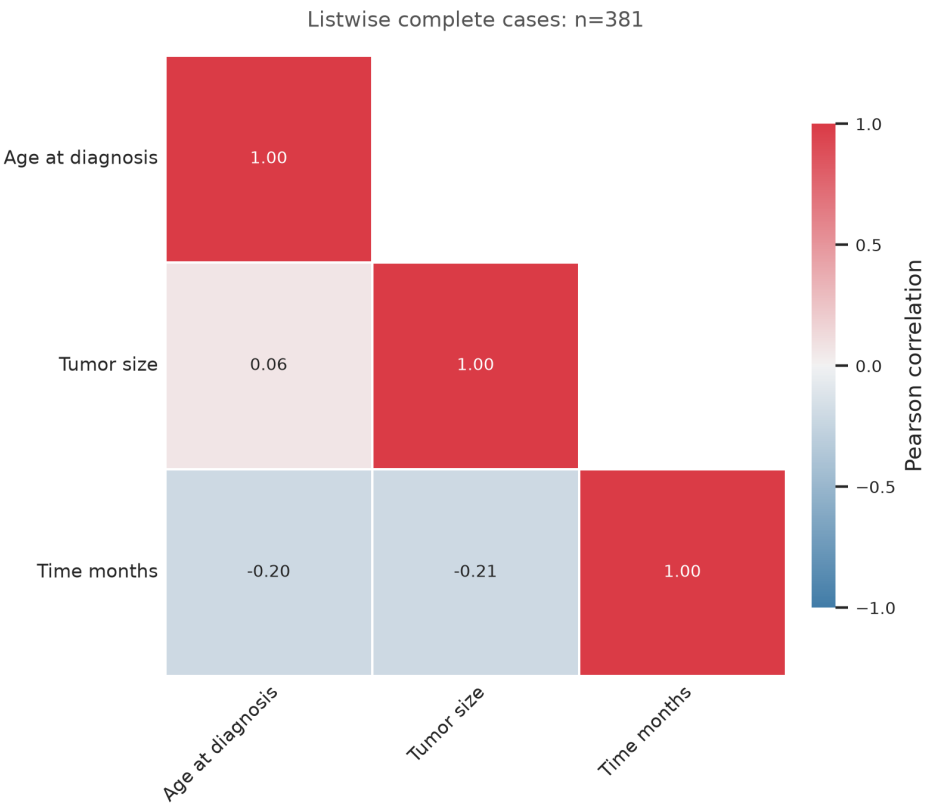
Agent analysis

Site And Federated Pearson — Site C Regional Hos



Human analysis

Regional Hospital: Local correlation heatmap



Values show Pearson correlation. Blank cells indicate undefined correlation, usually caused by a constant variable.

Human and agent correlation comparison - Federated

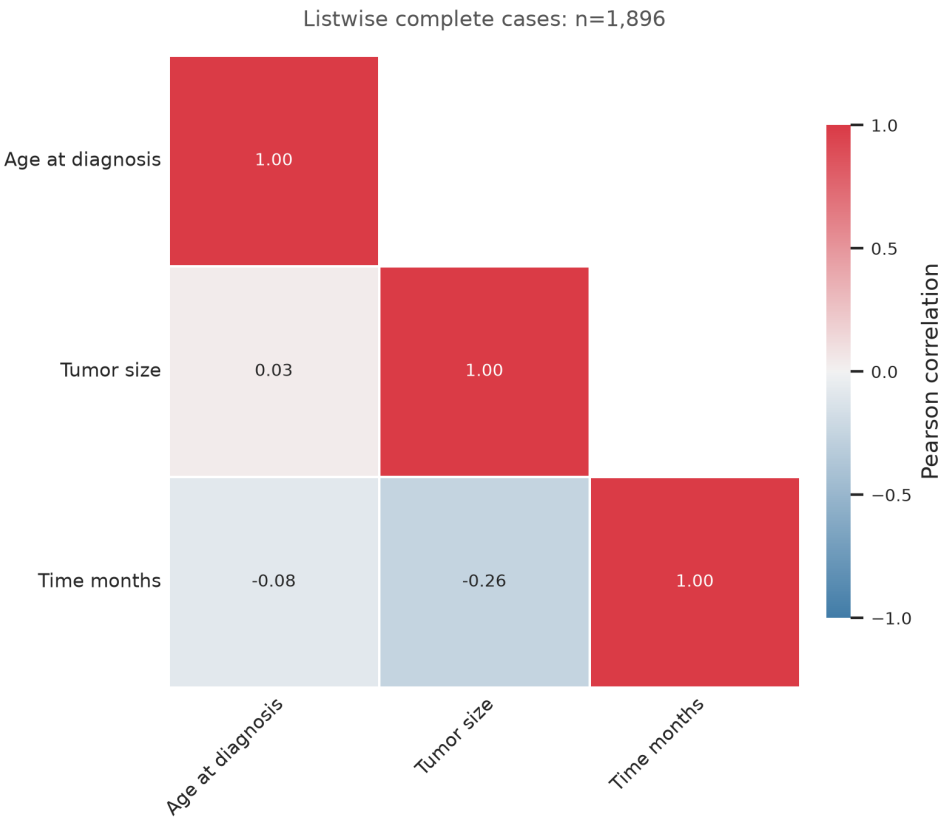
Agent analysis

Site And Federated Pearson — Federated



Human analysis

Federated METABRIC correlation heatmap



Values show Pearson correlation. Blank cells indicate undefined correlation, usually caused by a constant variable.

What the prototype demonstrates

- 01 Ask in natural language**

The agent converts an analysis question into a structured, inspectable plan.
- 02 Execute under governance**

NVFlare sends approved tasks to local sites while patient records remain local.
- 03 Reproduce the result**

Human and agent outputs can be compared with explicit numerical tolerances.